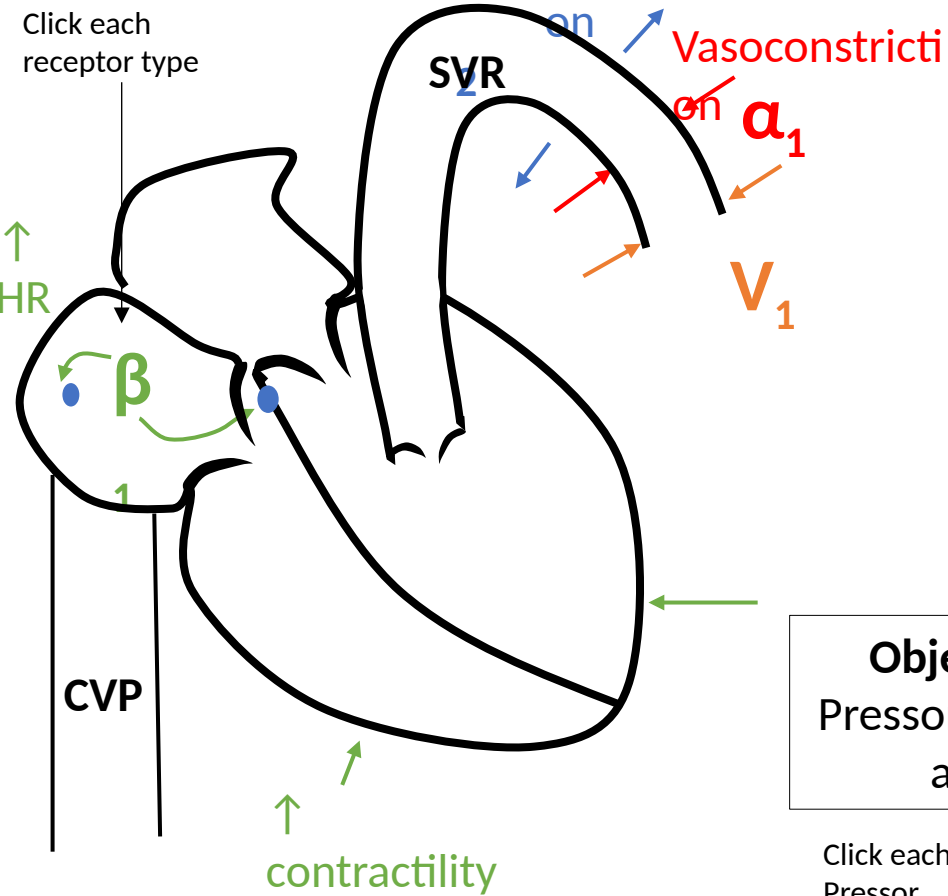


Objective 2:
CV receptor affects



Objective 3:
Pressor Affects and Uses

Click each Pressor →

Objective 1:
Physiology of Shock

Click each cause →

	Preload (CVP)	CO	Afterload (SVR)	Treatment
HYPOVOLEMIC	↓ ↓	↓	↑	Volume
CARDIOGENIC HF, valvular, arrhythmogenic	↑	↓ ↓	↑	Inotropes ± diuresis
OBSTRUCTIVE PE, PTX, tamponade	↑	↓	↑	Inotropes
DISTRIBUTIVE Sepsis Neurogenic Anaphylactic Adrenal crisis	↓ - nml	nml - ↑	↓	IVF + NE or vaso
Norepinephrine	-	mild ↑	↑ ↑	Distributive
Vasopressin	↑	-	↑	Distributive
Epinephrine	-	↑	↑	Anaphylaxis
Phenylephrine	-	-	↑	Distributive
Dobutamine	-	↑	mild ↓	Cardiogenic
Dopamine		↑	↓ → ↑	All

2021 Surviving Sepsis Campaign — what changed since this talk

Standing teaching is unchanged: norepinephrine remains the first-line vasopressor for septic shock.

Fluids: balanced crystalloids (LR/Plasma-Lyte) now suggested over 0.9% saline; the 30 mL/kg bolus within 3 h was downgraded from a strong to a weak recommendation.

Vasopressin: add vasopressin (rather than escalating norepinephrine) once NE reaches ~0.25–0.5 mcg/kg/min; combo was linked to lower mortality.

Start early, start peripheral: the guideline endorses initiating norepinephrine through a peripheral line to restore MAP rather than delaying for central access.

CLOVERS trial (NEJM 2023): no difference in 90-day mortality between a restrictive-fluid/early-pressor and a liberal-fluid strategy — both are acceptable.

Angiotensin II (ATHOS-3): FDA-approved add-on for refractory vasodilatory/distributive shock when catecholamines + vasopressin are insufficient.

Source: Evans L et al. Surviving Sepsis Campaign 2021, Crit Care Med 2021;49(11):e1063–e1143; National Heart Lung Blood Institute CLOVERS, NEJM 2023;388:499–510.

Take-Home Points

- Four broad categories of shock — hypovolemic, cardiogenic, obstructive, distributive — each with a distinct preload / cardiac-output / SVR profile that drives treatment.
- Inotropes (e.g., dobutamine) raise contractility for cardiogenic and obstructive shock; norepinephrine is the vasopressor of choice for distributive/septic shock.
- No vasopressors if "the tank is not full": hypovolemic shock is treated with volume (blood for hemorrhage); do not give IVF to cardiogenic shock.
- SvO₂ helps sort mixed pictures — low in cardiogenic shock, typically high in established septic shock (may be low in early sepsis); interpret alongside CO, SaO₂, and hemoglobin.
- 2021 update: balanced crystalloids over saline, add vasopressin early rather than max-dosing NE, and start NE peripherally to avoid delay.

Attribution & Optimization Notes

Adapted from TeachIM.org — "Shock and Pressors" (Yilin Zhang, MD; Brandon Fainstad, MD).

Original presenter board: Shock-and-Pressors-Presenter-Board-.pptx • Companion: Pressors-Learner-Handout.pdf.

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Optimization (June 2026): guideline currency verified — no factual errors found in the original board; teaching content is timeless physiology/pharmacology and remains current.

Added: 2021 Surviving Sepsis Campaign update slide (balanced crystalloids, early/peripheral NE, vasopressin, CLOVERS, angiotensin II), a take-home-points slide, and this attribution slide. All original graphics and click-to-reveal animations preserved.

Companion package: HTML lesson launcher, one-page clinical reference (PDF), Anki cards, ABIM-style MCQs, and an EMR dotphrase. Medical content should be clinician-reviewed before teaching.